Flash Tool/Web Terminal Flashing Firmware (Without IDF Development Environment)

Method 1: Flash software burning on computer (**without IDF Development Environment)**

Currently, the Xiaozhi AI firmware is applicable to development boards with ESP32 series main control chips. For details, please refer to the firmware download comparison table.

Note: 1)v0.X.X_ML307.zip is the 4G version firmware,**by defaultthe WiFi version should download v0.X.X_bread-compact-wifi.zip**(x.x represents the version number).

2) This document is an example of firmware burning operation. It is recommended to burn the latest firmware version during the burning process.

Install Flash Download Tool:https://www.espressif.com.cn/zh-hans/support/download/other-tools.

Or directly download the following Compressed Packet, unzip it, and you'll find the download tool inside:

flash_download_tool_3.9.7.zip
20.63MB

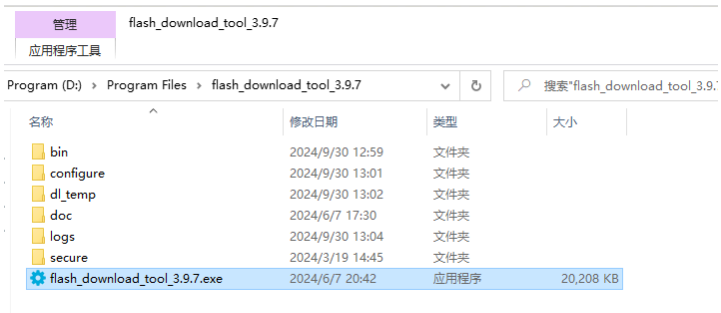
Connect the USB JTAG/serial of the ESP32 S3 development board to the computer host, run the Flash download tool, and use the UART mode to burn the decompressed merged-binary.bin to address 0x0.

1. Preparatory Work

On Windows systems,the currently recommended download/burning tool is Flash download_tool 3.9.7 version, and other versions can also be used as long as they are not too old.

After downloading, unzip flash_download_tool_3.9.7_1.zip to a directory of your choice on the hard drive. There is no need for installation. Simply enter the directory and double-click flash_download_tool_3.9.7.exe to run it.

Note that the programming tool should not be placed in a Chinese-named directory either , otherwise the file may not be loaded.



2. Download Firmware

(1) Download and extract

Click to open the firmware list: 🚀 Download the latest version firmware for Xiaozhi AI Terminal v1.5.5

Note: When storing the downloaded file on the local hard drive of my computer, **it is recommended that the path does not contain Chinese characters** , otherwise it may not be able to load and burn .

The following illustration is for reference only, **please download the latest firmware version when performing specific operations** .

Please select the corresponding version to download the firmware based on the development board and peripheral modules you are using. Download it to your computer's local hard drive.

Note: bread starts is the breadboard firmware (self-assembled by purchasing ESP32s3-WROOM-1/1U-N16R8 development board and modules, following the wiring scheme, using breadboard, DuPont wires, or self-welding). By default, the WIFI version is selected. If an ML307R (AT) 4G module is installed, please select the ML307 version.

Breadboard wifi product corresponds to: vx.x.x_bread-compact-wifi.zip

The 4G version of the breadboard corresponds to vx.x.x_bread-compact-ml307.zip (vx.x.x represents the version number, e.g., v1.5.5)

The Chinese display of the firmware corresponding to the development board/product is in pinyin or the corresponding English (please note) , please refer to the following figure:

Note: Please choose the corresponding firmware program for flashing yourself (vx.x.x represents the version number, e.g., v1.5.5). If the wrong program is selected, it may not run, and the screen will not display. For details about unfamiliar development boards, please consult the seller or search online.

序号	开发板/产品类型	固件BIN程序包（vx.x.x表示版本号例如 v1.5.5）
1	**面包板新版接线（WIFI）0.91OLED（44或42引脚esp32s3-wroom-1开发板）** BOARD_TYPE_BREAD_COMPACT_WIFI	vx.x.x_bread-compact-wifi.zip
2	**面包板新版接线（WIFI）+LCD（44或42引脚esp32s3-wroom-1开发板）** BOARD_TYPE_BREAD_COMPACT_WIFI_LCD	请自行根据LCD屏幕分辨率编译固件
3	**面包板新版接线（ML307 AT）** BOARD_TYPE_BREAD_COMPACT_ML307	vx.x.x_bread-compact-ml307.zip
4	**ESP BOX 3 （乐鑫）** BOARD_TYPE_ESP_BOX_3	vx.x.x_esp-box-3.zip
5	**Kevin Box 1** BOARD_TYPE_KEVIN_BOX_1	vx.x.x_kevin-box-1.zip
6	**Kevin Box 2** BOARD_TYPE_KEVIN_BOX_2	vx.x.x_kevin-box-2.zip
7	**Kevin C3** BOARD_TYPE_KEVIN_C3	vx.x.x_kevin-c3.zip
8	**Kevin SP V3开发板** BOARD_TYPE_KEVIN_SP_V3_DEV	vx.x.x_kevin-sp-v3.zip
9	**立创·实战派ESP32-S3开发板** BOARD_TYPE_LICHUANG_DEV	vx.x.x_lichuang-dev.zip
10	**立创·实战派ESP32-C3开发板** BOARD_TYPE_LICHUANG_C3	vx.x.x_lichuang-c3-dev.zip
11	**神奇按钮 Magiclick_2.4** BOARD_TYPE_MAGICLICK_2P4	vx.x.x_magiclick-2p4.zip
12	**神奇按钮 Magiclick_C3** BOARD_TYPE_MAGICLICK C3	vx.x.x_magiclick-c3.zip
13	**M5Stack CoreS3** BOARD_TYPE_M5STACK CORE S3	vx.x.x_m5stack-core-s3.zip
14	**AtomS3 + Echo Base** BOARD_TYPE_ATOMS3_ECHO_BASE	vx.x.x_atoms3-echo-base.zip
15	**AtomS3R + Echo Base** BOARDTYPE_ATOMS3R_ECHO_BASE	vx.x.x_atoms3r-echo-base.zip
16	**AtomMatrix+ Echo Base** BOARDTYPE ATOMMATRIX ECHO BASE	vx.x.x_atommatrix-echo-base.zip
17	**虾哥 Mini C3** BOARD TYPE_XMINI_C3	vx.x.x_xmini-c3.zip
18	**ESP32S3 KORV02 V3开发板** BOARDTYPEESP32S3_KORV02_V3	vx.x.x_esp32s3-korvo2-v3.zip
19	**ESP-SparkBot开发板（坦克）** BOARD_TYPE_ESP SPARKBOT	vx.x.x_esp-sparkbot.zip
20	**Waveshare ESP32-S3-Touch-AMOLED-1.8（微雪）** BOARD_TYPE_ESP32S3_Touch_AMOLED_1_8	vx.x.x_esp32-s3-touch-amoled-1.8.zip
21	**Waveshare ESP32-S3-Touch-LCD-1.85C** BOARD_TYPE_ESP32S3_Touch_LCD_1_85C	vx.x.x_esp32-s3-touch-lcd-1.85c.zip
22	**Waveshare ESP32-S3-Touch-LCD-1.85** BOARD_TYPE_ESP32S3_Touch_LCD_1_85	vx.x.x_esp32-s3-touch-lcd-1.85.zip
23	**Waveshare ESP32-S3-Touch-LCD-1.46** BOARD_TYPE_ESP32S3_Touch_LCD_1_46	vx.x.x_esp32-s3-touch-lcd-1.46.zip
24	**LILYG0 T-Circle-S3** BOARD TYPE LILYGOTCIRCLE S3	vx.x.x_lilygo-t-circle-s3.zip
25	**Movecall Moji 小智AI衍生版** BOARD TYPE MOVECALL MOJI ESP32S3	vx.x.x_movecall-moji-esp32s3.zip
26	**正点原子DNESP32S3开发板** BOARD_TYPE_ATK_DNESP32S3	vx.x.x_atk-dnesp32s3.zip
27	**正点原子DNESP32S3-BOX** BOARD_TYPE_ATK_DNESP32S3_BOX	vx.x.x_atk-dnesp32s3-box.zip
28	**面包板新版接线（WIFI）0.96OLED屏（44或42引脚esp32s3-wroom-1开发板）** BOARD_TYPE_BREAD_COMPACT_WIFI	vx.x.x_bread-compact-wifi-128x64.zip
29	**面包板新版接线（ML307 AT）0.96OLED屏（44或42引脚esp32s3-wroom-1开发板）** BOARD_TYPE_BREAD_COMPACT_ML307	vx.x.x_bread-compact-ml307-128x64.zip
30	**嘟嘟开发板CHATX(wifi)** BOARD_TYPE_DU_CHATX	vx.x.x_du-chatx.zip

31	**太极小派esp32s3** BOARD_TYPE_ESP32S3_Taiji_Pi	vx.x.x_taiji-pi-s3.zip
32	**无名科技星智0.96** BOARD_TYPE_XINGZHI_Cube_OLED	vx.x.x_xingzhi-cube-oled.zip
33	**无名科技星智1.54** BOARD_TYPE_XINGZHI_Cube_TFT	vx.x.x_xingzhi-cube-tft.zip

Note: For other development boards or products that do not have default support, directly flashing the program may not work. You need to modify the program source code, recompile it to adapt to the chip pin interfaces and related audio drivers, etc. (These operations require setting up the ESP-IDF Development Environment first and being familiar with the source code before modification and compilation, which will not be elaborated in this document.)

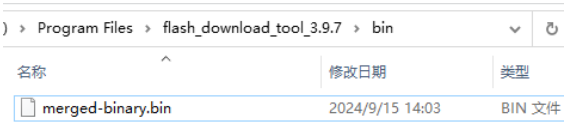
The following figure example selects the v1.5. 5 -bread-compact-wifi.zip version firmware, and after decompression, the **merged-binary.bin** file is obtained.

Latest firmware address: 🚀 Download the latest version of the firmware for the Xiaozhi AI Terminal

Use a breadboard to plug in the development board, and the default download is the WIFI version, while ML307 is the 4G version firmware.

(2) Copy the.bin file to the specified directory

Unzip the downloaded bin firmware compressed packet file, and it is recommended to place it in the bin directory of the flash download tool software. Please do not use Chinese directory names. As shown in the following figure:



(3) Others

Other release firmware versions and source code files are listed below. Please choose and download them according to your needs.

3. Burn Firmware / Download to Development Board

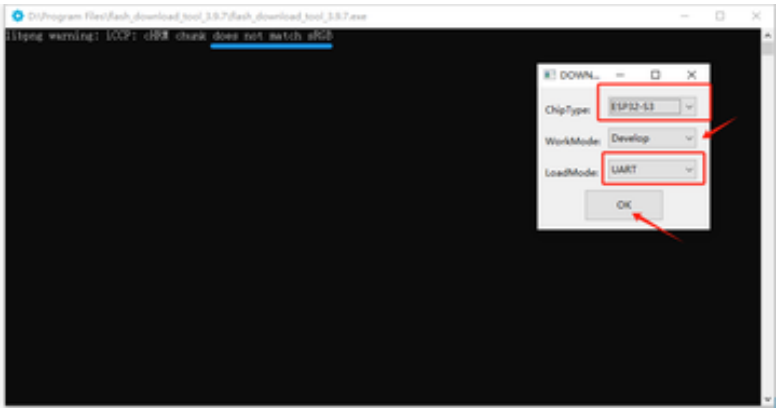
Navigate to the directory after extracting Flash download_tool_3.9.7, and double-click to run flash_download_tool_3.9.7.exe as shown in the following figure:

1) Download Settings

(1) Chip Type: For ESP32S3 development boards, please select ESP32-S3. If the user's development board uses other chips in the ESP32 series, please select the corresponding chip type ChipType. For example, for ESP32-C3 development boards/products, please select ESP32-C3.

(2) Working Mode, select Develop

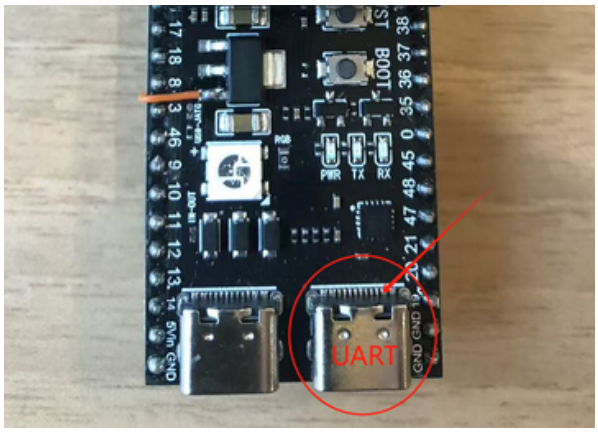
(3) For the loading mode, it is recommended to select UART (alternatively, USB mode is available but not described here)



Interface and sRGB etc. Instructions:

(1) When the Type-C interface on the front of the development board faces downward, the interface on the right is the UART interface (as shown in the figure below), and the one on the left is the USB interface. Please note not to connect them incorrectly.

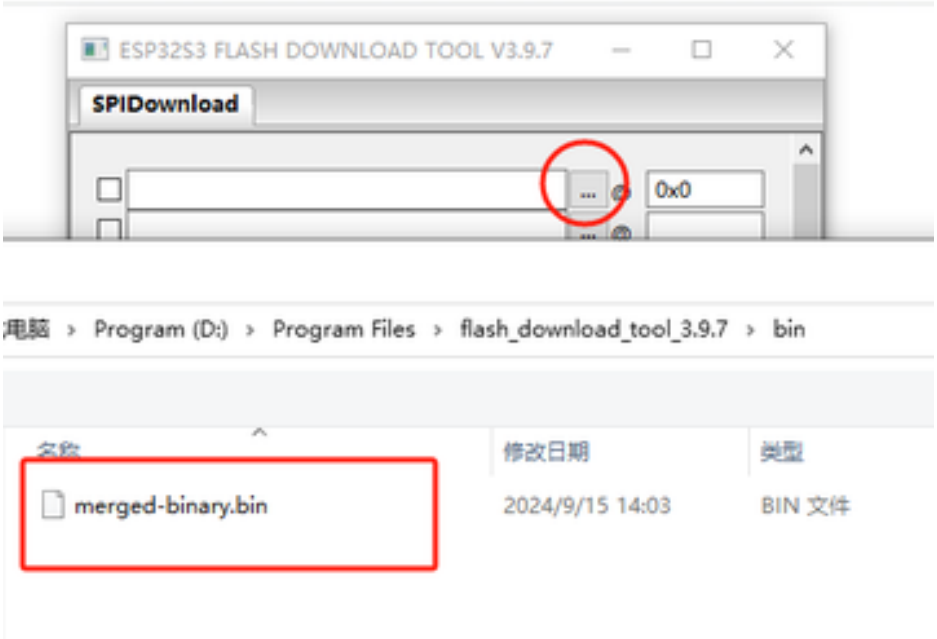
Supplementary note: Individual customized development boards may vary. Please identify the UART interface based on the silkscreen COM on the front or back of the development board's USB interface.



(2) If the switch of the sRGB light on your development board is not soldered (some manufacturers consider that some people may not need the sRGB light), the Flash download tool will fail to recognize it and issue a warning, which does not affect programming and can be ignored for now. You can then short-circuit it with an electric soldering iron (see position 2 in the illustration at the bottom).

2) Load Firmware & SPI Download Settings

(1) On the blank box in the first row... button, select the bin file directory to import the bin file, as shown in the following figure:



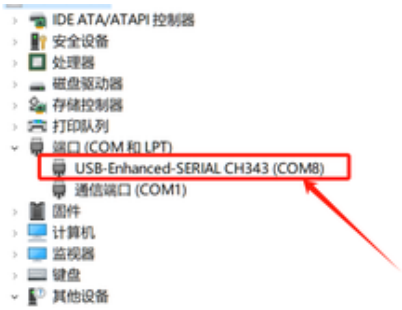
(2) Check the box in front of the bin file import selection box (it must be checked), and in the input box behind the bin file selection box, fill in 0x0 or 0x00 or 0 (indicating downloading to the starting position of memory area 1 of the development board)

Note: If you do not place the bin file in the bin directory of the flash download tool program but in other locations on your computer's hard drive, please do not use Chinese directory names, otherwise the programming tool will report an error when loading the bin file and programming will fail.

(3) COM Port: First, on the desktop, go to This PC -> Right-click menu Properties -> Device Manager (right-side menu) -> Ports (COM & LPT)**Click to expand**, check the COM Port Number corresponding to the inserted development board (if unsure, please plug and unplug the USB interface of the development board, and the COM Port Number that is uninstalled and newly recognized is the COM Port Number to be selected), which is the corresponding COM Port Number to be set in the FLASH DOWNLOAD TOOL during SPI download.

Note: The COM serial port of the ESP32S3 development board is generally **driven by CH340 or CH343** chips, with **CH34** in the port name, or in some cases, **CP210** and other chips, which are also UART ports.

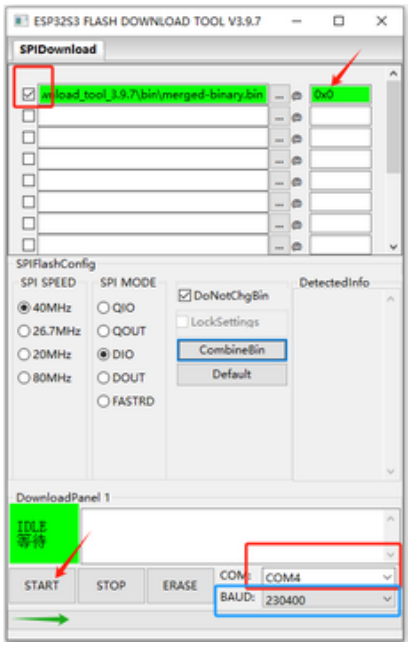
The COM8 port shown in the figure below is for reference only. Select the port of your development board based on the COM port displayed on your computer; it is not necessarily COM8, and should be determined by the COM port recognized by your computer.



(3) Others: SPI Speed SPEED and SPI Mode MODE can use the default settings (do not select QOUT and DOUT), and a higher BAUD rate can be selected.

** **Make sure to check the checkbox in front of the burned bin file!**

** **COM Port Number is selected according to the COM Port in your computer's Device Manager** (not necessarily COM4; the following figure is for demonstration only).



(4) Click START to begin the programming process. The Progress Bar will appear and continue until completion, with the status displaying FINISH upon completion. The download/programming process generally takes from a few minutes to about ten minutes. The following illustration shows the completed download:

(5) After the programming is completed, click the **RST** (Restart, as shown in Position 1 in the figure below) on the button to restart the development board, and then you can enter the WI-FI network configuration mode. For detailed information on network configuration, please refer to the next chapter.

Network Configuration and Device RegistrationLink: How to Configure Device Wi-Fi and Register Devices

Method 2: Third-party Xiaozhi dedicated firmware Web-based online flashing tool

1. Xiaozhi AI Firmware Dedicated Programming Tool - Ludaima IoT One-Stop Development Tool

Note: For a better experience, please use the Edge browser to access. Supports WeChat QR code login and account login.

https://dev.16302.com/tools/xiaozhiinit

The preview of the burning home page is as follows:

The dedicated programming tool for Xiaozhi firmware can burn various Xiaozhi operating firmware with one click. Its unique chip recognition function can automatically match the corresponding available firmware online. Using this tool, you can burn the newly released firmware in real time, eliminating the need for tedious environment setup work.

The main operating steps are as follows:

1.1 **Select Firmware**

Select cloud or local firmware

1.2 Connect the Device

Automatically identify or determine device type

1.3 Start Burning

Erase and burn firmware into the device

1.4 Completed

Flashing completed, prompting to disconnect the device. Return or enter the terminal to test the device.

Method 3: ESP-Launchpad Browser Web Terminal Flashing

1. Preparatory Work

1) Prepare the ESP32S3 development board to be programmed. For details, please refer to the supported development boards in the "Download Firmware" section of Method 1 above.

2) Connect the data cable of the development board (the standard data cable configured for mobile phones or development boards, etc., not the 2-core USB charging cable used for ordinary electronic products).

A desktop computer or laptop with a USB interface that can access the Internet.

3) Connect the ESP32S3 development board to the computer's USB port (some development boards need to turn on the power switch), and ensure that the development board has been recognized as a COM serial port in the computer - right-click Properties - Related Settings - Device Manager.

If the COM port is not found, please check the status of the data cable and the development board, and confirm that the serial bridge chip driver has been installed (the driver can be obtained from the seller or searched for online). If you are unsure which COM port it is, please refer to Section 3 of Method 1 for plug-and-unplug identification, etc.

4) Please use mainstream browsers, such as Google Chrome (verified), Microsoft Edge (verified) /IE browser, or other webkit-based browsers. Try not to use computer management software and antivirus software when flashing the firmware.

2. Download Firmware

Download the firmware of the corresponding version of Xiaozhi AI to the computer hard drive, extract the bin firmware file for later use.

Users of the breadboard-assembled Xiaozhi AI Terminal are defaulted to download **WiFi version v0.X.X_bread-compact-wifi.zip** (where x.x represents the version number).

Please do not use Chinese directories/paths. For specific operations, please refer to Section 2 of Method 1 in the above text.

Firmware Address: 🚀 Download the latest version of the Xiaozhi AI Terminal firmware v1.5.5

3. Burn Firmware / Download to Development Board

1) **Access the burning URL via a browser**

Online programming URL:https://espressif.github.io/esp-launchpad/

If you see there is no follow-up, it may be that the administrator is editing. First, click on the link above to do some research. Or wait until I finish writing.

The burning website used in this tutorial is in English. The homepage contains an introduction to the function of burning application firmware. Note that burning new firmware will overwrite the original firmware. The default setting on the website's Quick Start page is to burn the application firmware of ESP-RainMaker, so there's no need to worry about it. **We'll directly switch to the DIY menu page.**

2) DIY page connects to the COM serial port of the Connect development board

Operating Steps:

① Please click on the **DIY page** in the upper right corner of the website **Connect menu** (arrow position 1).

② On the **pop-up layer** , **select the COM serial port of the development board connected to the user's computer** , as shown in the arrow position 2 in the figure below (USB Serial (COM12) - Paired). The COM port number here is for reference only and may vary depending on each person's development board.

③ On **the pop-up layer** , **click the Connect button for authorization** , so that the user's development board COM serial port is loaded by the website browser.

④ Confirm that the COM serial port of the development board is connected

After the development board's COM port is connected by the browser, the web page will display **Connected to device: ESP32-S3** , and at the same time, **the menu Connect** will change to **Disconnect, indicating that the development board's COM port is successfully connected.**

3) Load and set the corresponding version of the BIN firmware

1) The webpage loads the corresponding bin firmware

Click the "Select File" button (Position 1 in the figure below) in the middle of the DIY page, browse to the folder where the bin firmware is located on your computer, select the bin firmware file (Position 2 in the figure below), then click the "Open" button (Position 3 in the figure below) to load the bin firmware onto the webpage.

2) Set the starting position of programming

Operating Steps:

Please fill in the input box for Flash Address (the starting address of the flash storage area) with 0x0 or 0 , as shown in Position 1 in the figure below.

Meanwhile, please ensure that the bin file just loaded is visible at position 2. If it is not displayed, please re-import it according to step 1).

Since the Xiaozhi AI firmware package has been merged into a single bin package, namely merged-binary.bin, there is no need to add additional bin firmware.

4) Burn the bin firmware and check if the burning is successful

(1) Start the programming process:

Please click the Program button at position 3 in the above illustration to start burning the firmware program. A prompt will indicate that the corresponding application firmware will be downloaded to the development board/device.

(2) Burn Process Display

After starting the burn process, the web page will **automatically jump to the Console command-line menu page** , displaying the burn progress, as shown in the following figure:

(3) Confirm the programming is completed and restart

- **Programming completed**

After seeing the "Write Complete 100%" and the prompt message "Leaving..." on the Console black screen, click the "Restart Device" button on the right to restart the development board, and you will see that the program has been successfully flashed.

- **Restart the device/development board**

After the firmware is flashed, the device will not restart automatically. You need to manually restart the device. Here **please click the Reset Device button on the web page** (Position 1 in the figure below) **, and confirm again by clicking the Confirm button** (Position 2 in the figure below) **to confirm the restart of the development board** to start running the firmware program just flashed.

- **The firmware program of the development board is running normally**

After the development board/device is restarted, the serial port log of the Xiaozhi AI terminal firmware program will be loaded on the Console. The appearance of the network card address and WIFI network configuration prompt in the output log indicates that the program has been successfully flashed, as shown in the following figure.

4. Error Messages and Exceptions

Q1: Why can't the development board connect?

The development board cannot connect to the browser, the pop-up layer options are empty, and it cannot be loaded and displayed on the web page. Consider that the interface may be plugged in incorrectly, or the wrong option may be selected when there are multiple ports, or the COM serial port of the development board may be occupied by other software on the computer.

Q2: Why can't the Program Burn button be clicked?

The bin file has been loaded, but the "Burn Program Firmware" button is grayed out and cannot be clicked.

1) Generally, check whether the development board has been connected to the computer browser, and display the words " **Connected to device: ESP32-S3** " on the webpage.

2) When the browser switches pages, in order to speed up the display, there will be a cache. Please click the browser menu to refresh the page again, or press F5 or Ctrl + R to refresh. After refreshing, if the selected file is not loaded, please reload it, or disconnect the development board **Disconnect** and reconnect it, and then check whether the program button appears. If there is still a problem, you can try another browser.

Q3: Why does the program not run properly after programming?

After programming, the program continuously restarts on the black screen of the Console, or an E-level error occurs, and it cannot work properly. Considering that the user's development board does not support the currently programmed firmware, please confirm whether the firmware for the corresponding development board is correct. Please refer to the "Download Firmware" section in Method 1.

For more exceptional issues not mentioned above, please go to the QQ Group for discussion.

QA Issues and**Others**

1. **How to enter boot mode?**

Hold down the BOOT key, then press the RST key (for about 0.5 seconds), release the BOOT key to set the board into programming mode. You can try it repeatedly, and the serial port will prompt "Programming Mode".

Auto program

DTR	RTS->EN	IO0	
1	1	1	1
0	0	1	1
1	0	0	1
0	1	1	0

Attachment: Download Q Part Drivers for ESP32S3 Development Board

Download link for Qinheng USB to Serial Port Driver, thanks to the group members for collection and sharing!

CH340 Series:

https://www.wch.cn/products/CH340.html

CH343 Series:

https://www.wch.cn/products/CH343.html